

LAB SESSION 3

Programming Exercise:

Question 1: Define a structure named Person with members name and age. Create a variable of this structure type, initialize it, and display its contents.

```
C:\C++ PROGRAMS > g++ tut1.cpp > main()
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 // Define the structure
6 struct Person
7 {
8     string name;
9     int age;
10 };
11
12 int main()
13 {
14     // Create and initialize a variable of type Person
15     Person person1;
16     person1.name = "Ali";
17     person1.age = 20;
18
19     // Display the contents
20     cout << "Name: " << person1.name << endl;
21     cout << "Age: " << person1.age << endl;
22
23     return 0;
24 }
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Name: Ali
Age: 20
PS C:\C++ PROGRAMS>
```

Question 2: Define a structure named Rectangle with members length and width. Create a variable of this structure type, input values for length and width, calculate the area, and display it.

```
C:\C++ PROGRAMS > g++ tut1.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 // Define the structure
5 struct Rectangle
6 {
7     float length;
8     float width;
9 };
10
11 int main()
12 {
13     Rectangle rect;
14
15     // Input values
16     cout << "Enter the length of the rectangle: ";
17     cin >> rect.length;
18
19     cout << "Enter the width of the rectangle: ";
20     cin >> rect.width;
21
22     // Calculate area
23     float area = rect.length * rect.width;
24
25     // Display the area
26     cout << "The area of the rectangle is: " << area << endl;
27
28     return 0;
29 }
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter the length of the rectangle: 5
Enter the width of the rectangle: 3
The area of the rectangle is: 15
PS C:\C++ PROGRAMS>
```

Question 3: Define a structure named **Number** with a single member **num**. Create two variables of this structure type, input values for them, swap their values using a function, and display the swapped values.

```
#include <iostream>
using namespace std;

// Define the structure
struct Number {
    int num;
};

// Function to swap values of two Number variables
void swapNumbers(Number &a, Number &b) {
    int temp = a.num;
    a.num = b.num;
    b.num = temp;
}

int main() {
    Number n1, n2;

    // Input values
    cout << "Enter the first number: ";
    cin >> n1.num;

    cout << "Enter the second number: ";
    cin >> n2.num;

    // Display original values
    cout << "\nBefore swapping:\n";
    cout << "n1 = " << n1.num << ", n2 = " << n2.num <<
endl;
    // Swap the values
    swapNumbers(n1, n2);

    // Display swapped values
    cout << "\nAfter swapping:\n";
    cout << "n1 = " << n1.num << ", n2 = " << n2.num <<
endl;
    return 0;
}
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter the first number: 20
Enter the second number: 40

Before swapping:
n1 = 20, n2 = 40

After swapping:
n1 = 40, n2 = 20
PS C:\C++ PROGRAMS> █
```

Question 4: Define a structure named Student with members rollNumber, name, age, and marks. Create an array of 3 Student structures, input data for each student, display their details, and find the student with the highest marks.

```
Enter details for Student 1:
Roll Number: 12
Name: Ali
Age: 20
Marks: 88

Enter details for Student 2:
Roll Number: 15
Name: Ahmed
Age: 21
Marks: 98

Enter details for Student 3:
Roll Number: 14
Name: Abid
Age: 23
Marks: 78

--- Student Details ---

Student 1:
Roll Number: 12
Name: Ali
Age: 20
Marks: 88

Student 2:
Roll Number: 15
Name: Ahmed
Age: 21
Marks: 98

Student 3:
Roll Number: 14
Name: Abid
Age: 23
Marks: 78

--- Student with Highest Marks ---
Roll Number: 15
Name: Ahmed
Age: 21
Marks: 98
PS C:\C++ PROGRAMS> █
```

```

#include <iostream>
using namespace std;

// Define the structure
struct Student {
    int rollNumber;
    string name;
    int age;
    float marks;
};

int main() {
    Student students[3];

    // Input data for 3 students
    for (int i = 0; i < 3; i++) {
        cout << "\nEnter details for Student " << i + 1 << ":\n";
        cout << "Roll Number: ";
        cin >> students[i].rollNumber;
        cout << "Name: ";
        cin.ignore(); // To clear input buffer before getline
        getline(cin, students[i].name);
        cout << "Age: ";
        cin >> students[i].age;
        cout << "Marks: ";
        cin >> students[i].marks;
    }

    // Display details of all students
    cout << "\n--- Student Details ---\n";
    for (int i = 0; i < 3; i++) {
        cout << "\nStudent " << i + 1 << ":\n";
        cout << "Roll Number: " << students[i].rollNumber << endl;
        cout << "Name: " << students[i].name << endl;
        cout << "Age: " << students[i].age << endl;
        cout << "Marks: " << students[i].marks << endl;
    }

    // Find the student with the highest marks
    int topIndex = 0;
    for (int i = 1; i < 3; i++) {
        if (students[i].marks > students[topIndex].marks) {
            topIndex = i;
        }
    }

    // Display the student with highest marks
    cout << "\n--- Student with Highest Marks ---\n";
    cout << "Roll Number: " << students[topIndex].rollNumber << endl;
    cout << "Name: " << students[topIndex].name << endl;
    cout << "Age: " << students[topIndex].age << endl;
    cout << "Marks: " << students[topIndex].marks << endl;

    return 0;
}

```

Question 5: Define a structure named Employee with members employeeID, name, department, and salary. Create an array of 5 Employee structures, input data for each employee, display their details, and calculate the average salary of all employees.

```
Employee ID: 03
Name: Ahmed
Department: EL
Salary: 670000

Enter details for Employee 4:
Employee ID: 04
Name: Sana
Department: EMD
Salary: 350000

Enter details for Employee 5:
Employee ID: Sarah
Name: Department: Salary:
--- Employee Details ---

Employee 1:
ID: 1
Name: Ali
Department: SE
Salary: 50000

Employee 2:
ID: 2
Name: Abid
Department: CSIT
Salary: 60000

Employee 3:
ID: 3
Name: Ahmed
Department: EL
Salary: 670000

Employee 4:
ID: 4
Name: Sana
Department: EMD
Salary: 350000

Employee 5:
ID: 0
Name:
Department:
Salary: 3.50325e-44

Average Salary of Employees: 226000
PS C:\C++ PROGRAMS> █
```

```

#include <iostream>
using namespace std;

// Define the structure
struct Employee {
    int employeeID;
    string name;
    string department;
    float salary;
};

int main() {
    Employee employees[5];
    float totalSalary = 0;

    // Input data for 5 employees
    for (int i = 0; i < 5; i++) {
        cout << "\nEnter details for Employee " << i + 1 << ":\n";
        cout << "Employee ID: ";
        cin >> employees[i].employeeID;
        cin.ignore(); // Clear input buffer
        cout << "Name: ";
        getline(cin, employees[i].name);
        cout << "Department: ";
        getline(cin, employees[i].department);
        cout << "Salary: ";
        cin >> employees[i].salary;

        totalSalary += employees[i].salary; // Add salary to total
    }

    // Display employee details
    cout << "\n--- Employee Details ---\n";
    for (int i = 0; i < 5; i++) {
        cout << "\nEmployee " << i + 1 << ":\n";
        cout << "ID: " << employees[i].employeeID << endl;
        cout << "Name: " << employees[i].name << endl;
        cout << "Department: " << employees[i].department << endl;
        cout << "Salary: " << employees[i].salary << endl;
    }

    // Calculate and display average salary
    float averageSalary = totalSalary / 5;
    cout << "\nAverage Salary of Employees: " << averageSalary <<
endl;
    return 0;
}

```

Question 6: Define a structure named Book with members title, author, genre, and yearOfPublication. Create an array of 3 Book structures, input data for each book, display their details, and find the book with the earliest publication year.

Enter details for Book 1:

Title: C++ Basics

Author: Bjarne Stroustrup

Genre: Programming

Year of Publication: 1985

...

--- Book with the Earliest Publication Year ---

Title: C++ Basics

Author: Bjarne Stroustrup

Genre: Programming

Year: 1985

```

#include <iostream>
using namespace std;

// Define the structure
struct Book {
    string title;
    string author;
    string genre;
    int yearOfPublication;
};

int main() {
    Book books[3];

    // Input data for 3 books
    for (int i = 0; i < 3; i++) {
        cout << "\nEnter details for Book " << i + 1 << ":\n";
        cout << "Title: ";
        cin.ignore(); // Clear input buffer
        getline(cin, books[i].title);
        cout << "Author: ";
        getline(cin, books[i].author);
        cout << "Genre: ";
        getline(cin, books[i].genre);
        cout << "Year of Publication: ";
        cin >> books[i].yearOfPublication;
    }

    // Display book details
    cout << "\n--- Book Details ---\n";
    for (int i = 0; i < 3; i++) {
        cout << "\nBook " << i + 1 << ":\n";
        cout << "Title: " << books[i].title << endl;
        cout << "Author: " << books[i].author << endl;
        cout << "Genre: " << books[i].genre << endl;
        cout << "Year of Publication: " << books[i].yearOfPublication << endl;
    }

    // Find the book with the earliest publication year
    int earliestIndex = 0;
    for (int i = 1; i < 3; i++) {
        if (books[i].yearOfPublication < books[earliestIndex].yearOfPublication)
        {
            earliestIndex = i;
        }
    }

    // Display the book with the earliest publication year
    cout << "\n--- Book with the Earliest Publication Year ---\n";
    cout << "Title: " << books[earliestIndex].title << endl;
    cout << "Author: " << books[earliestIndex].author << endl;
    cout << "Genre: " << books[earliestIndex].genre << endl;
    cout << "Year: " << books[earliestIndex].yearOfPublication << endl;

    return 0;
}

```

Question 7: Define a structure named Product with members productName, price, and quantity. Create a variable of this structure type, input data for the product, calculate the total cost, and display it.

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1
```

```
Enter Product Name: Notebook
```

```
Enter Price: 50
```

```
Enter Quantity: 3
```

```
--- Product Details ---
```

```
Product Name: Notebook
```

```
Price: 50
```

```
Quantity: 3
```

```
Total Cost: 150
```

```
PS C:\C++ PROGRAMS> █
```

```
#include <iostream>
using namespace std;

// Define the structure
struct Product {
    string productName;
    float price;
    int quantity;
};

int main() {
    Product item;

    // Input data
    cout << "Enter Product Name: ";
    getline(cin, item.productName);
    cout << "Enter Price: ";
    cin >> item.price;
    cout << "Enter Quantity: ";
    cin >> item.quantity;

    // Calculate total cost
    float totalCost = item.price * item.quantity;

    // Display product details and total cost
    cout << "\n--- Product Details ---\n";
    cout << "Product Name: " << item.productName << endl;
    cout << "Price: " << item.price << endl;
    cout << "Quantity: " << item.quantity << endl;
    cout << "Total Cost: " << totalCost << endl;

    return 0;
}
```

